

20 A

$$\begin{aligned} R \text{ of inner core} &= \rho l / A \\ &= (1.7 \times 10^{-8}) (10) / [\pi \times (0.0050 / 2)^2] \\ &= 0.00866 \, \Omega \end{aligned}$$

$$\begin{aligned} R \text{ of outer layer} &= \rho l / A \\ &= (2.7 \times 10^{-8}) (10) / [\pi \times (0.010 / 2)^2 - \pi \times (0.0050 / 2)^2] \\ &= 0.00459 \, \Omega \end{aligned}$$

Since the inner core and the outer layer are in parallel,

$$\begin{aligned} 1 / R_{\text{total}} &= 1 / 0.00866 + 1 / 0.00459 \\ R_{\text{total}} &= 0.0030 \, \Omega \end{aligned}$$